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SUPERSEDING
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FEDERAL SPECIFICATION

BLEACHERS, PORTABLE

This specification was approved by the Assistant Administrator, Office of Personal Property, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers prefabricated modular and beam type portable bleachers, skid and trailer mounted.

1.2 Classification.

1.2.1 Types. Bleachers shall be one of the following types, styles, and sizes as specified (see 6.2 and figures 1 and 2 for illustrations).

Type I - Modular construction.
Type II- Beam construction.

1.2.2 Styles (see 6.2).

Style A - Skid mounted. The skid mounted bleacher shall be of types I or II.

Style B - Trailer mounted. The trailer mounted bleacher shall be of types I or II or a combination of one base section of five rows of seats and two, three or four foldaown sections of four rows of seats.

1.2.3 Size. Unless otherwise specified (see 6.2), bleacher units shall have one of the following seating capacities:

Style A:

Size 60 - Not less than 60 seat spaces. (5 rows only)
Size 120 - Not less than 120 seat spaces. (10 rows only)

Style B:

Size 120 - Not less than 120 seat spaces.
Size 180 - Not less than 180 seat spaces.
Size 240 - Not less than 240 seat spaces.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Commercial Item Description (CID):

A-A-880 - Strapping, Steel, Flat and Seals.

Federal Specifications:

MM-L-751 - Lumber; Softwood.
TT-W-550 - Wood Preservative: Chromated Copper Arsenate Mixture
TT-W-00571 - Wood Preservation; Treating Practices
TT-W-572 - Wood Preservative, Water-Repellent.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Box, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

FSC 5450

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(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-T-704 - Treatment and Painting of Material.
MIL-A-8625 - Anodic Coating, for Aluminum and Aluminum Alloys.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, Trailer and Trailer Dollies; Preservation and Packaging of.

(Copies of Military specifications, and standards, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations (CFR):

49 CFR - Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

American Society for Testing and Materials (ASTM) Standards:

A36 - Structural Steel.
A242 - High Strength Low Alloy Structural Steel.
A307 - Carbon Steel Externally Threaded Standard Fasteners.
B221 - Aluminum-Alloy Extruded Bars, Rods, Wire Shape and Tube.
B308 - Aluminum-Alloy Standard Structural Shapes, Rolled or Extruded.
B316 - Aluminum-Alloy Rivet and Cold-Heading Wire and Rods.
B429 - Aluminum-Alloy Extruded Structural Pipe and Tube.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American Wood Preservers Association (AWPA):

Standard C-2 Lumber, Timber, Bridge & Mine Ties, Pressure Treatment of.
Standard C-9 Water Borne Preservatives
Standard P-5 Plywood, Pressure Treatment

(Application for copies should be addressed to the American Wood Preservers Association, 7735 Old Georgetown Road, Bethesda, MD 20814.)

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Building Officials and Code Administrators International Inc. (BOCA):

BOCA - The BOCA Basic Building Code.

(Application for copies should be addressed to the Building Officials and Code Administrators International Inc., 1313 E. 60th Street, Chicago, IL 60637.)

National Fire Prevention Association (NFPA):

NFPA 102 - Chapter 8, Grandstand and Bleachers

NFPA 102 Chapter 9, Folding and Telescopic Seating.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Society of Automotive Engineers, Inc., (SAE):

SAE J560 - Seven-Conductor Electrical. Connector for Truck-Trailer Jumper Cable
SAE J534 - Lubrication Fittings, Standard For

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

Southern Pine Inspection Bureau:

Standard Grading Rules for Southern Pine Lumber.

Application for copies should be addressed to the Southern Pine Inspection Bureau, P.O. Box 846, Pensacola, FL 32502.)

Tire and Rim Association Inc.:

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

West Coast Lumber Inspection Bureau:

Standard Grading Rules for West Coast Lumber.

(Application for copies should be addressed to the West Coast Lumber Inspection Bureau, P.O. Box 15406, Portland, OR 97225.)

3. REQUIREMENTS

3.1 Description. The bleachers shall be of modular or beam construction, skid or trailer mounted. The bleacher(s) hereinafter shall be referred to as unit(s).

3.2 First article. When specified (see 6.2), the contractor shall furnish a complete unit with accessories for first article inspection and approval (see 4.2.1 and 6.4).

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3.3 Standard commercial product. The unit(s) shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified. Unit understructure shall be of painted steel, galvanized steel, or aluminum.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts (see 4.6.1).

3.6 Seating area and supporting frame materials. All materials that come in contact with each other shall be compatible to prevent galvanic corrosion.

3.6.1 Structural steel. Steel for structural members shall conform to ASTM A36 or A242.

3.6.2 Tubular steel. Steel tubing for structural members shall be structural grade, high carbon, steel tubing; yield strength not less than 50,000 pounds per square inch (psi); tensile strength not less than 75,000 psi; elongation in 2 inches, 15 percent. Tubing of alloy steel may be used provided the strength and corrosion resistance are equivalent or superior.

3.6.3 Aluminum. Aluminum for structural members, seatboards, and footboards shall conform to ASTM B221, B308, B316, or B429 as required to meet the design parameters. Anodizing shall be in accordance with MIL-A-8625 in either clear or gold color.

3.6.4 Lumber. (1) Lumber shall be Southern Pine or Douglas Fir. Lumber for seatboards and footboards shall be kiln dried to a moisture content not exceeding 19 percent. Lumber shall be surfaced four sides in accordance with MM-L-751 and shall be legibly marked "TOP" by indentation in the wood of the one side which has a pitch-free face. Lumber shall be in accordance with MM-L-751 and meet specification standards of one of the following:

(a) No. 1 Stadium Grade or No. 1 Dense Stadium Grade according to Standard Grading rules for Southern Pine Lumber; or
(b) Stadium Plank according to Standard Grading Rules of West Coast Lumber or .

(2) Lumber shall be pressure treated, meeting AWPA Standard P-5 and Fed. Spec. TT-W-550. The treatment shall be applied in a closed cylinder by vacuum-pressure process in accordance with the approved standards and recommended treating practice as listed in AWPA Standards C-2 and C-9 or the appropriate AWPA Standard covering the commodity treated and listed in Fed. Spec. TT-W-571.

3.7 Design and construction. The units, excluding the trailer mounted, shall be furnished unassembled, complete with seatboards, footboards, and supporting framework as specified herein. The seating capacity of the unit shall be based on a width per seat of not less than 18 inches nor more than 19 inches. Modular construction, type I, shall have a supporting framework of 3, 4, or 5 seat row modules, and the multiple row modules are joined to form the full depth of the unit for the required capacity. Beam construction, type II, shall have support beams or stringers extending full depth of the unit at right angles at the seat rows. Supporting modular frames or stringers shall be spaced not more than six (6) feet centertocenter. Supporting members shall be cross-braced throughout the entire structure in a manner to adequately meet the design load requirement. Supporting framework may be galvanized or painted steel or aluminum. All fasteners, straps, yokes, and plates shall be furnished complete by the contractor. The units shall be regular commercial items which meet the following minimum requirements. See figures 1 and 2 for clarity. Design and configuration is not restricted to illustrations, but are offered as general examples only. The trailer mounted unit, (style B) shall be basically the same as the skid-mounted unit (style A) with the exception of fold-down sections and the trailer. The style B base section and 5 rows of seats shall be permanently attached to the trailer frame. The 2, 3, or 4 fold-down sections (depending on size) of 4 rows of seats, shall be attached to the base section on a pivot for each

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section. When the sections are folded down in the seating mode and the trailer stabilized, the unit shall be ready for use. The support section and seating section for all units shall otherwise conform to NFPA 102 and BOCA requirements for grandstands and bleachers and other requirements herein.

(a) Height to top of first seat:	not to exceed 18 inches.
(b) Depth between seat rows:	not less than 22 inches nor more than 24 inches.
(c) Rise per row:	not less than 8 inches, nor more than 10 inches (style A) nor more than 11 inches (style B)
(d) Vertical distance from footboard to corresponding board:	not less than 16 inches, nor more than 18 inches.
(e) Live load per linear foot of seatboard and of footboard:	not less than 120 pounds (lb)
(f) Horizontal sway load, applied lengthwise to seats, per linear foot:	not less than 24 lb.
(g) Horizontal sway load, applied perpendicular to seats, per linear foot:	not less than 10 lb.
(h) Guardrails at ends and back; horizontal load at top rail, per linear foot:	not less than 20 lb.
(i) Vertical load on top rail, per linear foot:	not less than 75 lb.

Guardrails shall be furnished on ends and back of bleacher at locations where top of seatboards are more than 4 feet above grade (see 3.7.8). Unless otherwise specified (see 6.2), sleepers or sills for bolting to the base of the framework or supports shall be furnished. All materials in contact shall be compatible to prevent galvanic corrosion.

3.7.1 Bolts and nuts. Bolts and nuts shall be zinc or cadmium coated steel or stainless steel and shall conform to ASTM A307, or shall be aluminum conforming to ASTM B316.

3.7.2 Flat washers. Flat washers shall be commercial grade and resistant to atmospheric corrosion.

3.7.3 Lock washers. Lock washers shall be either aluminum, stainless steel, or commercial grade zinc coated steel.

3.7.4 Framework. The framework shall be fabricated from steel or aluminum structural shapes, tubing, or pipe as specified (see 6.2). Structural members shall be shop drilled or punched and reamed for field connections and fasteners. Tubing members shall be welded at the seam.

3.7.5 Seatboards and footboards (wood). Seatboards and footboards (nominal 2 inch by 10 inch planking) as specified (see 6.2), shall be cut to design length and jig drilled and relieved for connection to the supporting structure. Seatboards shall butt flush and have edges eased. Exposed surfaces shall be free from slivers and cleaned, treated, and painted after all machining operations. Pressurized wood shall not be painted. Seatboards and footboards shall not deflect more than 1/2 inch between supports under maximum allowable seating load calculated at 120 lb live load per linear foot over a 6 foot span, under a 2-span condition.

3.7.6 Seatboards and footboards (aluminum). Aluminum seatboards and footboards as specified (see 6.2), shall be typical nominal 2 inch by 10 inch extruded plank designed to carry a load of 120 lb per linear foot over a 6-foot span, under a 2-span condition, with a deflection when fully loaded of not more than 1/2 inch. Seatboards shall be anodized satin etched finish in silver or gold. Footboards shall be nonanodized natural mill finish. Both seatboards and footboards shall be attached to bleacher understructure by means of hotdipped galvanized bolts, nuts, lockwashers and mill finished aluminum bolt clips. Ends of all aluminum planks shall be covered with suitable end caps of 0.050-inch (minimum) aluminum sheet, formed to fit snugly over the end of each plank and be attached by means of aluminum blind rivets. End caps shall have a finish matching that of seatboards and footboards.

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3.7.7 Ground sills. Ground sills or sleepers shall be wood, steel, or aluminum bolted to the bleacher framework. The lumber (except pressurized wood) treated in accordance with TT-W-572, using either pentachlorophenol or water-borne preservatives for structural timbers, paintable, suitable for contact with the ground. Sills or sleepers shall be cleaned and painted after all machine operations. Metal sills or sleepers shall be primed and painted or zinc coated for steel. Aluminum shall be anodized conforming to MIL-A-8625, to prevent corrosion. Sills or sleepers for style A (skid mounted), shall be of size and strength to support the calculated dead and live load, without exceeding 1/2 inch deflection between weight bearing members when the bearing soil consists of in-place gravel and sand gravel mixtures. Sill and sleeper ground bearing strength in this paragraph does not pertain to the trailer-mounted unit(s) (style B). Sills and sleepers shall otherwise conform to NFPA 102, chapter 8.8.

3.7.8 Guardrail. Guardrails shall be of welded or bolted structural steel or aluminum, steel or aluminum angle, steel or aluminum pipe or tubing with threaded, slip-on, key-clamped or welded connections. Rails shall be bolted or welded to uprights. Uprights shall be bolted or welded to the basic structure. Bolts shall be provided with locking devices to prevent loss of tightness. All surfaces and connections shall be ground smooth, corners chamfered and free from snag causing features. All materials in contact shall be compatible to prevent galvanic corrosion. Guardrails shall otherwise conform to NFPA 102, chapters 8 and 9 for grandstand and bleachers, and folding and telescopic seating.

3.7.8.1 Style A bleacher guardrail spacing. Rails shall be spaced and extend to a height as follows (see 3.7.8.1.1 and 3.7.8.1.2):

3.7.8.1.1 Endrails height and spacing.

Top surface of top rail	- 42 inches minimum from center of top surface of seatboard.
Bottom rail	- Space between bottom rail and top rear edge of footboard shall not permit a sphere larger than 9 inches in diameter to pass through.
Intermediate rails	- Space between toprail, intermediate rail and bottom rail shall not permit a sphere larger than 9 inches in diameter to pass through.

3.7.8.1.2 Back rails. Back rails shall have same spacing and height as endrails and shall not permit a sphere larger than 9 inches in diameter to pass between the bottom rail and the top rear edge of tne rear seatboard.

3.7.8.2 Style B bleacher guardrails. Rails shall be spaced and extend to a height as follows (see 3.7.8.2.1 and 3.7.8.2.2):

3.7.8.2.1 Endrails height and spacing.

Top surface of top rail	- 36 inches minimum from center of top surface of seatboard
Bottom rail	- Space between bottom rail and top rear edge of footboard shall not permit a sphere larger than 9 inches in diameter to pass through.
Intermediate rails	- Space between top rail, intermediate rail and bottom rail shall not permit a sphere larger than 9 inches in diameter to pass through.

3.7.8.2.2 Back rails. Back rails shall have same height and spacing as endrails and shall not permit a sphere larger than 9 inches in diameter to pass between the bottom rail and the top rear edge of the rear seatboard.

3.7.8.3 Removable rails. When specified (see 6.2), the guardrails of style B shall be removable or side rails shall telescope into storage for traveling, shipping, or storage. Provisions shall be provided for securing the guardrails to the unit when removable for traveling, shipping, and storage.

3.8 Folddown winch. The style B unit shall be provided with manually operated winch(s) for raising and lowering the folddown sections. The winches shall be easily operated by one man for this operation. The winch shall be secured to a vertical support (boom or stantion) to provide the required height for the winch to allow for the proper angle between the winch line and the folding section. The winch line shall be of adequate working load strength with a 5:1 safety factor. The support with winch shall be positioned in a pocket on the supporting frame or trailer frame. Removable pins shall be provided to secure the support to prevent circular and vertical movement. The support shall be removed while the unit is to be used for seating and for traveling. Provisions shall be provided for securing the winch and support to the unit when removed for traveling and when the unit is used for seating, or an aluminum torsion bar system shall be utilized enabling two (2) men to handle each sectionalized seating module.

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3.9 Instruction manual. One set of instruction manuals, including drawings or diagrams and bill of material, shall be furnished by the contractor with each unit, clearly indicating the location of each piece by appropriate designation. Erection instructions shall include, but is not limited to: precautions concerning overloading; methods of blocking, leveling, anchoring; maintenance and repair instructions; critical inspection points and recommended inspection intervals (see 6.3).

3.10 Trailer. The trailer for style B units shall, as a minimum, conform to specifications as follows:

3.10.1 General design. The trailer shall comply with 49 CFR for trailers, as applicable, in effect at time of manufacture. The trailer shall be similar, but not restricted, to the illustration in figure 2, and shall be suitable for highway use. All materials in contact shall be compatible to prevent galvanic corrosion.

3.10.2 Net weight. The net weight shall consist of the trailer, added accessories, the seating section and supporting frame including guardrails, jacks, booms or torsion bar system (As applicable).

3.10.3 Rated payload capacity. The rated payload capacity, with the payload uniformly distributed over the seating area, shall be calculated as not less than 120 lb multiplied by the total linear feet of seating area.

3.10.4 Gross weight. The gross vehicle weight shall consist of the net weight, and a uniformly distributed live load of not less than 100 lb per square foot multiplied by the gross horizontal projection.

3.10.5 Trailer dimensions and net weight. The dimensions and net weight shall be as shown in table I in transport mode.

TABLE I. Dimensions and net weight.

Size	Maximum length	Maximum width	Maximum weight (lb)
120 seats	24 feet	8 feet	4,500
180 seats	34 feet	8 feet	5,500
240 seats	44 feet	8 feet	7,000

3.11 Performance. The trailer, fully equipped, shall be capable of being towed at speeds as great as 5 miles per hour (mph) over unimproved roads and over reasonably hard uneven terrain; and at speeds as great as 55 mph over improved roads.

3.11.1 Turning ability. When coupled to a towing vehicle operating in its minimum turning circle, the trailer shall follow without cramping. The trailer shall be capable of moving through an angle of not less than 70° in relation to a 96-inch wide towing vehicle without interference between vehicles. The angle shall be measured between the longitudinal axis of the towing vehicle and the couple device and towbar centerline.

3.11.2 Tracking ability. The trailer shall conform to the tracking requirements of 49 CFR, section 393.70(a).

3.11.3 Brake performance. The trailer, furnished with electric or hydraulic surge braking system shall comply with the performance requirements of 49 CFR, section 393.52.

3.12 Trailer components.

3.12.1 Frame and towbar. The frame and towbar shall be of all steel construction and shall have structural capacity to resist bending and twisting, without permanent deformation or part failure, when subjected to performance requirements of 3.11 through 3.11.2 When the trailer is loaded with rated payload, the maximum fiber stress in any frame member shall be not greater than 50 percent of the yield strength of the materials used. When frame members are spliced, joints shall be reinforced to fulfill the dynamic and static loading requirements. All splice welds shall be continuous, and splices shall be designed to avoid stress concentration. Towbar shall conform to 49 CFR, section 393.71(h).

3.12.2 Coupling device. As specified, a lunette, or ball and socket, shall be provided (see 6.2). Coupling device shall conform to 49 CFR, section 393.70(d).

3.12.3 Safety chains. Safety chains, conforming to the requirements of 49 CFR, section 393.70(d), and SAE J697 shall be furnished.

3.12.4 Support legs.

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3.12.4.1 Trailer towbar support leg. Unless otherwise specified (see 6.2), a support leg shall be mounted on the trailer towbar. The leg shall be vertically adjustable, hinged to fold into travel position, and shall be arranged for positive locking into parking or travel position. Leg shall be capable of raising or lowering its portion of the weight of the trailer, throughout the full extension without evidence of binding or other damage. Extension mechanism shall be operated by means of a handcrank. Leg adjusting mechanism shall be protected from exposure to foreign matter which would impair functional efficiency. The adjustments shall permit coupling and uncoupling the trailer, to or from the towing vehicle without the use of auxiliary equipment. Leg shall be provided with substantial ground bearing plate. Provisions shall be provided for locking the handcrank, with a padlock, to prevent unauthorized tampering with the support leg adjustment.

3.12.4.2 Trailer frame support legs. Unless otherwise specified (see 6.2), a minimum of four (4) legs shall be provided to support the complete unit while in the seating mode. Legs shall be adjustable to allow for leveling, 6 inch minimum, and retract for traveling. Legs shall be designed to support the gross weight (see 3.10.4) when the wheels and draw bar support leg is clear of the ground. Legs shall be attached to the trailer frame and of adequate strength to resist the forces when tested (see 4.6.2.6). Legs shall be provided with adequate bearing plates to prevent penetration in excess of 1 inch in bearing soil indicated in 3.7.6 As a minimum, one leg shall be provided at or near each corner, and one midway at each side (when the distance between corners is over 30 feet 1 inch), of the trailer frame.

3.12.5 Fenders. Unless otherwise specified (see 6.2), wheel fenders shall be provided. Fenders shall have sufficient clearance above the tires to provide for wheel articulation as specified in 3.12.6.1 Wheel fenders shall have sufficient structural strength to withstand loads of 200 lb per square foot without evidence of permanent deformation. A flange or lip shall be provided on the outside of the fenders.

3.12.6 Axles and suspension. Axle and suspension system shall be of rated capacity at least equal to the load imposed at the ground when the trailer is net loaded (see 3.10.2). Wheel bearings shall be of antifriction type. Trailers shall be provided with a tandem axle.

3.12.6.1 Tandem axle. Tandem axle suspension shall be individually suspended or walking beam construction, which will provide positive alignment. Sufficient articulation shall be provided to maintain equal loading on each wheel, with trailer loaded with rated payload, under any of the following conditions:

(a) With two diagonally opposite wheels elevated 5 inches.
(b) With forward pair of wheels elevated 6 inches above rear pair.
(c) With rear pair of wheels elevated 6 inches above forward pair.

3.12.7 Wheels, rims, and tires. Unless otherwise specified (see 6.2), trailers shall be equipped with single wheels. Tire and rim sizes shall be the same for all wheels on the trailer.

3.12.7.1 Tires. Unless otherwise specified (see 6.2), tires shall be tube or tubeless type with highway tread. Tires shall have individual rated load-carrying capacities at least equal to the maximum individual tire loading imposed in operation with net payload (see 3.10.2). Tires shall be rated in accordance with T&RA recommendations.

3.12.7.2 Tubes. When tube type tires are furnished, tubes shall be of heavy-duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires as recommended by T&RA.

3.12.8 Brakes. Brakes shall conform to 49 CFR, sections 393.40 through 393.43. No part of the braking system shall extend below the bottom of the wheel rim. Brakes shall be furnished on all wheels of the trailer. Trailer shall have electric or hydraulic surge type service brakes.

3.12.8.1 Electric service brakes. Electric service brakes shall be complete with brake controller operable from the driver's compartment of the towing vehicle. The electric service brakes shall be 12-volt direct current (dc) potential and connection with the towing vehicle shall be through the SAE J560 receptacle specified in 3.12.9. The kits shall be complete with all accessory parts and shall be boxed, marked, and securely attached to the trailer. A complete set of servicing and installation instructions shall be provided.

3.12.8.2 Hydraulic surge brakes. Actuator system shall be totally self contained in the hitch and steel understructure.

3.12.8.3 Emergency brake system. Electric or hydraulic surge type shall be provided with standard breakaway feature that will automatically apply the trailer brakes if trailer separates from towing vehicle conforming to 49 CFR, section 393.43.

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3.12.9 Lighting system. The lighting system shall be manufacturer's standard of 12-volt dc potential and shall comply with 49 CFR, sections 393.14, 393.20, 393.22, 393.25 through 393.29, 393.32, and 393.33. The system shall include at least two stoplights and taillights, directional signals, front and rear marker and clearance lights, and reflector devices. The front of the trailer shall be equipped with a receptacle conforming to SAE J560, with the receptacle and its conductors connected and color coded as specified therein. The receptacle shall be furnished with a spring loaded cover assembly and installed on the forward end of the drawbar. When specified, the rear light bar shall be removable (see 6.2).

3.12.9.1 Electrical jumper cable. When specified (see 6.2), a 7-conductor electrical jumper cable conforming to SAE J1067, with a cable plug at each end conforming to SAE J560, shall be provided. Both plugs shall be equipped with a grip for withdrawing from connector sockets. The cable shall be not less than 3 feet longer than the distance from the trailer electrical receptacle to the coupling device.

3.13 Servicing and adjusting. Prior to acceptance of the trailer by the Government inspector, the contractor shall service and adjust the trailer for immediate operational use. The servicing and adjusting shall include at least the following: Inflation of all tires, adjustment of brakes, proper functional of all lighting, and complete lubrication.

3.13.1 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be properly lubricated before delivery and shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.14 Workmanship.

3.14.1 Fabrication. Material used in the fabrication of components shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Re-entrant cuts shall be made in the best possible manner. Steel may be flame cut, using a tip suitable for the thickness of the metal, instead of sheared or sawed. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove slag and cracks. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. Bolt holes shall be accurately punched or drilled and shall have the burrs removed.

3.14.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. All weld spatter and projections on mating surfaces where members are in contact for fabrication or assembly shall be ground to a suitable flat surface. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Manual and machine welding processes and materials shall be in accordance with a nationally recognized welding code for the purpose intended.

3.14.3 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.14.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.14.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.15 Cleaning, treatment, and painting. All surfaces, except zinc-coated and threaded surfaces or aluminum surfaces, shall be cleaned, treated, and painted in conformance with MIL-T-704, type A. Finish color shall be as specified (see 6.2). Aluminum seatboards shall be treated as specified in 3.6.3 unless otherwise specified (see 6.2).

3.16 Instruction plates. The unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter.

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3.17 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor shall stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly and/or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8 inch in diameter shall be used to affix the plates. Nomenclature shall be: "BLEACHERS, PORTABLE (TYPE OF BASE-SKID OR TRAILER MOUNTED), (SEATING CAPACITY)".

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirement specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).
(c) Preparation for delivery inspection (see 4.7).

4.2.1 First article inspection. When first article is required (see 3.2 and 6.2), first article inspection shall be performed on the first production item. This inspection shall include the examination of 4.5 and the tests of 4.6 through 4.7. The first article may be a standard production item from the supplier's current inventory provided the unit meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining units to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on the sample units selected in accordance with 4.4. This inspection shall include the examination of 4.5 and the tests of 4.6.1 through 4.6.2.7.

4.3 Inspection lot. All units of the same type, style, and size as applicable, offered to the Government at one time, shall be considered a lot for purposes of inspection. The sample unit shall be one complete unit.

4.4 Sampling. A random sample of units shall be selected from each lot in accordance with MIL-STD-105.

4.4.1 Sampling for examination. Examination of the units shall be based on inspection level S-2 and an Acceptance Quality Level (AQL) of 6.5 percent defective.

4.4.2 Sampling for tests Test of the units shall be based on inspection level S2 and AQL of 4.0 percent defective.

4.5 Examination. Each unit selected for inspection shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy or suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.6 Tests.

4.6.1 Erection test. A preproduction sample of a completed unit shall be selected as a test erection unit, and shall be erected on a level base. The parts shall fit together without modification, forcing, or racking. Interchangeability shall be determined by transposing corresponding members (see 3.5). The contractor shall provide all labor, facilities, and supplies necessary for assembly and disassembly, and shall repair, including touchup painting, or replace any parts damaged as a result of this test. The erection test unit may be used as a completed production unit.

4.6.2 Load tests.

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4.6.2.1 Seatboards and footboards. Seat and footboards shall be loaded with a uniformly distributed load of not less than 120 lb per linear foot over a 6-foot span in a two-span condition (see 3.7(e)). Deflection shall be measured and shall not exceed 1/2 inch. The load shall be removed and deflection checked again to assure there is no permanent set or deformation to the planks.

4.6.2.2 Horizontal sway load - lengthwise to seats. With unit erected, resting on a firm even ground, all ground contact support points butted against block to prevent horizontal movement lengthwise, apply a uniformly distributed horizontal force (of 24 lb per linear foot of all seats) (see 3.7(f)), at one end of the unit at a point where the seat boards rest.

Example of force:
1.5 feet per seat
120 seating capacity
(120) (1.5) (24 lb) =
(180 linear feet) (24 lb) = 4,320 lb.

Release the force and inspect for broken bolts, welds, or deformed members.
Cracked or broken welds, broken bolts, or deformed members shall constitute failure of the test.

4.6.2.3 Horizontal sway load - perpendicular to the seats. With the unit positioned as in 4.6.2.2 and blocked to prevent horizontal movement, apply a uniformly distributed force of 10 lb per linear foot of seats, perpendicular to the seats, horizontally, to each row of seats, either from front or rear of seatboards (10 lbs/ft x length of seatboard.)

Example of force:
120 seat capacity (10 rows)
=(10#/ft x 18 ft/row x 10 rows)
= 1800 lbs force.

Release the force and inspect as in 4.6.2.2. Cracked or broken welds, broken bolts, or deformed members shall constitute failure of the test.

4.6.2.4 Guardrail horizontal force. A force equal to 20 lb per linear foot of rail shall be applied, horizontally, to each guardrail section (back, each end) separately (see 3.7(h)).

Example:
Backrail 40 feet long
(40) (20) = 800 lb force.

Release the force and inspect. Cracked or broken welds, broken bolts, joints or couplings, or deformed members shall constitute failure of the test.

4.6.2.5 Guardrail vertical force. A force equal to 75 lb per linear foot of rail, in separate sections (back, each end) shall be applied, uniformly distributed, vertically downward (see 3.7(i)) to the rail. The force shall be applied as a free swinging suspended load or other force without lateral support. A bumper block may be attached to the sloped rail to prevent the load from sliding down.

Example of load:
Rail - 40 feet long
(40) (75) = 3,000 lb.

Remove the force and inspect. Cracked or broken welds, broken bolts, joints or couplings, or deformed members shall constitute failure of the test.

4.6.2.6 Trailer mounted unit, style B, load test. The style B unit shall be tested as prescribed in 4.6.1 through 4.6.2.5, in the seating mode, completely assembled. During testing, the entire unit shall be resting on the support legs (tires and towbar support off the ground).

4.6.2.7 Trailer mounted unit set-up. The unit shall be setup, from towing mode, within 60 minutes by one man. With the unit on even level ground and coupled to a tow vehicle, the unit shall be completely uncoupled, support legs lowered and locked. The folddown sections lowered, guardrails installed (if applicable), hand winch(es) and support removed, and towbar support locked. The unit shall be ready for occupancy (seating mode) within 60 minutes.

4.7 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

5. PREPARATION FOR DELIVERY

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5.1 Preservation-packaging. The preservation-packaging shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. The fasteners, bolts, nuts, and washers shall be packaged in accordance with MIL-P-116, method III, in boxes conforming to PPP-B-636, class weather-resistant. Items furnished in commercial packages shall be overwrapped with waterproof barrier material with all folds, seams, and laps sealed with adhesive.

5.1.1.1 Technical publications. Technical publications for each unit shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.2 Level B. The fasteners, bolts, nuts, and washers shall be preserved and packaged as specified in 5.1, except that the boxes shall conform to PPP-B-636, class domestic, and the barrier wrap shall not be required. Closure of boxes shall be in accordance with the appendix to PPP-B-636.

5.1.3 Commercial. The bleachers shall be packaged in accordance with the normal commercial practice. The complete package shall be designed to protect the bleachers against damage during shipment, handling, and storage.

5.2 Packing. The packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The bolts, nuts, washers, and any other small components not suitable for bundling shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type. The boxed items shall be secured to the bundles capable of multiple shipments handling, and prolonged storage. For units unassembled, the structural members, guardrails, ground sills, seatboards, and footboards shall be bundled into secured lifts with flat steel strapping and bolting in combination with suitable wood blocking or battens, as required to form nonshifting bundles. The strapping shall conform to A-A-880, regular duty, grade optional, zinc finish, and be spaced not to exceed 36 inches on center, with end strapping placed not more than 18 inches from each end. The strapping shall be stapled to any wood blocking or battens provided to prevent movement during shipment and handling.

5.2.2 Level B. The unit shall be packed as specified for level A, except that the boxes shall conform to PPP-B-636, class weather-resistant, style optional. Closure, waterproof sealing, and strapping of the boxes shall be in accordance with the appendix to PPP-B-636.

5.2.3 Commercial. The packaged bleachers shall be packed to insure delivery at destination, provide for redistribution by the initial receiving activity, and be acceptable by common carrier under the National Motor Freight Classification rules and Uniform Freight Classification rules.

5.2.4 Style B units. When specified (see 6.2), style B units shall be preserved and packaged in accordance with 5.2.4.1 or 5.2.4.2.

5.2.4.1 Civil agencies. Unless otherwise specified, the unit shall be preserved and packaged in accordance with the supplier's standard practice.

5.2.4.2 Military agencies. Unless otherwise specified, the units shall be preserved and packaged in accordance with the (mobile) requirements of MIL-STD-281, with the level of preservation-packaging as specified by the procurement activity.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages, shipping containers, and bundles shall conform to Fed. Std. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages, shipping containers, and bundles shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The units are used to seat personnel, with a seating capacity of 60, 120, 180, or 240.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type of unit required (see 1.2.1).
(c) Style of unit required (see 1.2.2).

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(d) Seating capacity required (see 1.2.3).
(e) When first article inspection is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
(f) Aluminum treatment required (see 3.6.3 and 3.15).
(g) When different lumber is required (see 3.6.4).
(h) When sleepers or sills are not required (see 3.7.7).
(i) Type of frame material (see 3.7.4).
(j) Type of seatboards and footboards (see 3.7.5 and 3.7.6).
(k) Removable rails required (see 3.7.8.3).
(l) Type of couple device required (see 3.12.2).
(m) When towbar support not required (see 3.12.4.1).
(n) Trailer frame support legs required (see 3.12.4.2).
(o) When fenders not required (see 3.12.5).
(p) Wheel requirements, if different (see 3.12.7).
(q) Tire requirements, if different (see 3.12.7.1).
(r) When removable light bar required (see 3.12.9).
(s) When electrical jumper cable required (see 3.12.9.1).
(t) Lubrication requirement if different (see 3.13.1).
(u) Color of finish paint (see 3.15).
(v) Level of preservation-packaging, and packing required (see 5.1, 5.2 and 5.2.4).
(w) Whether marking shall be in accordance with Fed. Std. 123 or MIL-STD-129 (see 5.3).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 16.815 of the Defense Acquisition Regulations (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 16.815 are not invoked, the Deliverable data required by this specification cited in the following paragraph:

Paragraph	Data Requirement	Applicable DD 1664
3.9	Manual: Erection/Assembly/Maintenance	UDI-M-24026

(Copies of Data Item Descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, tests, and approval of the first article.

MILITARY INTERESTS:

Custodians

Army - GL
Navy - YD

Review Activity

DLA - CS

User Activity

Navy - MC

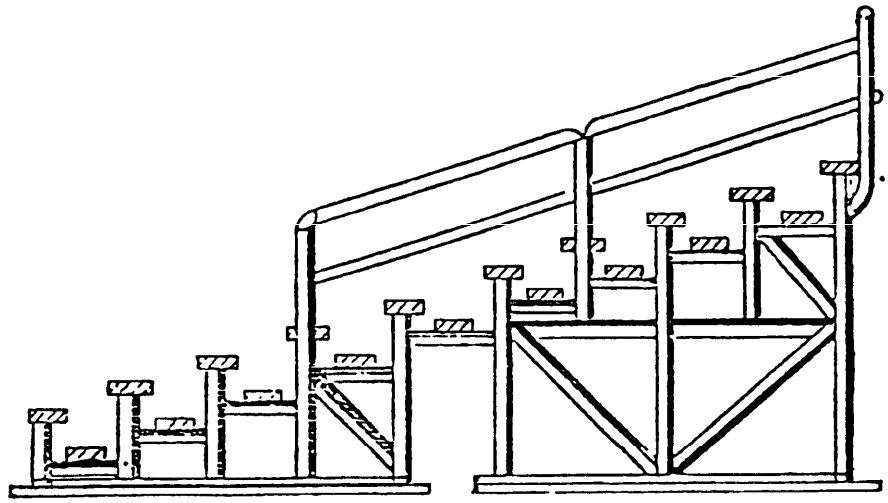
CIVIL AGENCY COORDINATING ACTIVITIES:

DC GOVT - DCG
HHS - FEC
INTERIOR - BLM
JUSTICE - FPI
USDA - AFS

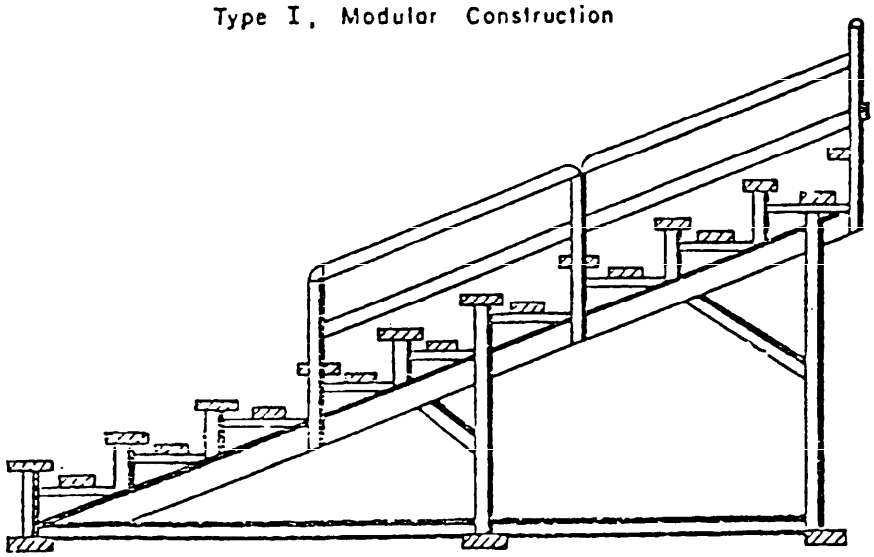
PREPARING ACTIVITY:

GSA - FSS

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Type I, Modular Construction



Type II, Beam Construction

FIGURE 1. Portable bleachers-end view, style A.

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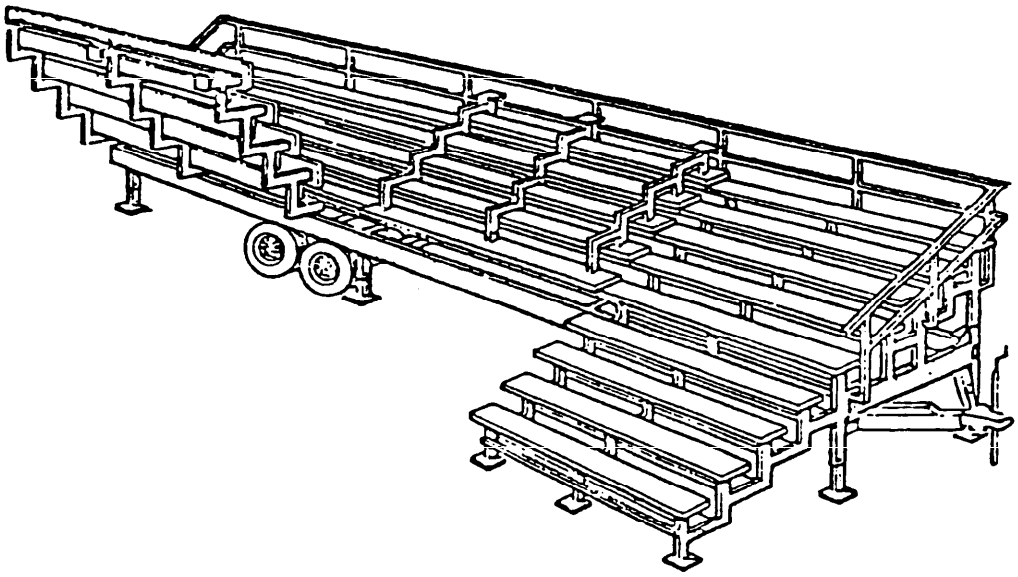


FIGURE 2. Portable bleachers, style B - trailer mounted.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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